MA860H Stepper Motor Driver

一、Overview

MA860H type sub-type two-phase hybrid stepping motor driver, using AC18-80V or DC24 ~ 110V power supply for 8.0A drive current is less than the outer diameter of 57 to 86 mm of two-phase hybrid stepping motor. This drive AC servo drive current loop subdivision control, small torque ripple of the motor, smooth running at low speed, low vibration and noise. When a relatively high-speed output torque, high positioning accuracy. Widely used in engraving machines, CNC machine tools, packaging machinery, transmission equipment and other requirements of the higher resolution devices.

Main features

1 Average current control, two-phase sinusoidal current drive output

2 AC18-80V or DC24 ~ 110V power supply

3 optically isolated signal input / output

4 overvoltage, undervoltage, overcurrent, short circuit protection and white

5 fourteen file segmentation and automatic half current function

6 eight file output phase current setting

7 with offline functionality

8 high start speed

9 high speed torque

Electrical parameters

Input voltage	AC18-80V or DC24 ~ 110V power supply
Input current	less than 8 A
Output Current	2.0A～7.8A
Temperature	Operating temperature -10 ~ 45 ℃; Storage temperature -40 ℃ ~ 70 ℃
Wet	No condensation and water droplets
Gas	Prohibit combustible gases and conductive dust
Weight	450g

二、Control signal interface

Figure 1 is a schematic wiring drive

1, the control signal definition

PUL +: positive pulse signal input terminal

PUL-: negative pulse signal input terminal

DIR +: Stepping positive direction signal input terminal

DIR-: Stepping negative direction signal input terminal

ENA +: Offline enable reset signal input positive terminal

ENA-: Offline enable reset signal input negative terminal

Offline enable signal is reset when a valid drive failure, prohibit any valid pulse, the output power of the drive element is closed, no motor holding torque.

2,control signals connection

PC control signal can be active high, you can also active low. When active high, all the negative terminal of the control signal as the signal ground together, when active low, all the positive terminal of the control signals together as a signal common terminal.

Now with open collector output and PNP for example, the interface circuit diagram is as follows:

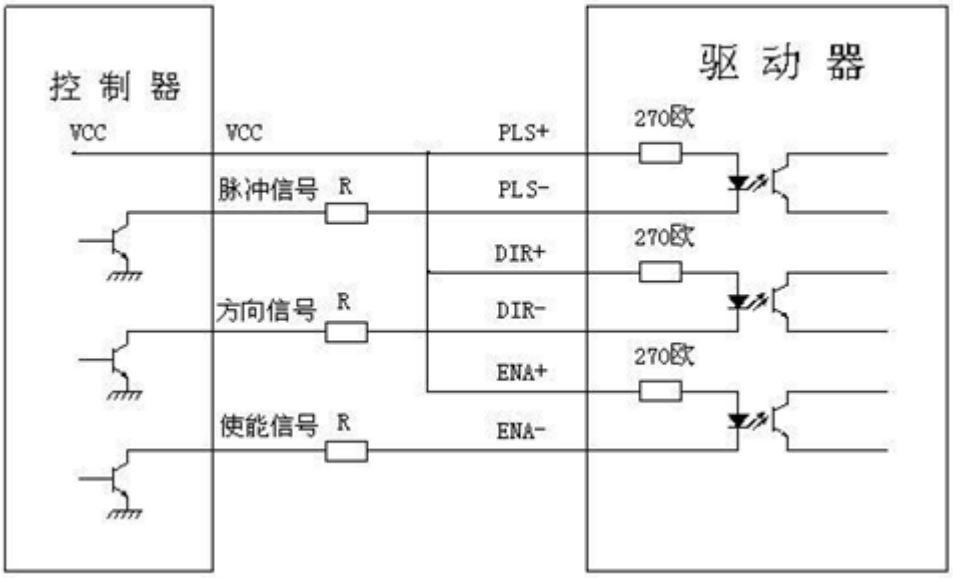


Figure 1. Input interface circuit (common anode connection)
Controller Open collector output

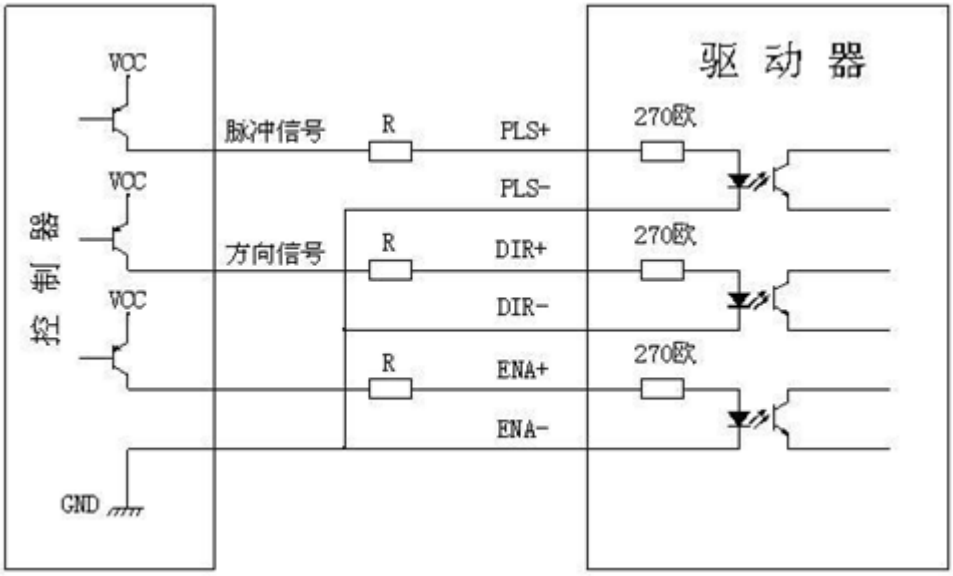


Figure 2. Input interface circuit (common cathode connection)
Controller PNP output

Note: VCC is 5V, R is short;
VCC is 12V, R is 1K, 1/8W greater resistance;
VCC is 24V, R is 2K, more than 1/8W resistor
R must be connected to the controller signals

三、Function Select (implemented by DIP switches on the drive panel)

1, set the number of steps per motor revolution

Drive motor may be a few steps per revolution are set to 400,800,1000,1600,2000,3200,5000,6400,10000,12800,25000,25600,50000,51200 step. SW5, SW6, SW7, SW8 bit to set the number of steps drives users to drive through the front panel DIP switches (see Table 1):

SW5	on	off	on	off	on	off	on	off	on	off	on	off	on	off
SW6	on	on	off	off	on	on	off	off	on	on	off	off	on	on
SW7	on	on	on	on	off	off	off	off	on	on	on	on	off	off
SW8	on	on	on	on	on	on	on	on	off	off	off	off	off	off
Pulse/rev	400	800	1600	3200	6400	12800	25600	51200	1000	2000	5000	10000	25000	50000

2, the control mode selection

DIP switch SW4 bit can be set to two control modes:

When set to "OFF", in order to have a semi-streaming capabilities.
When set to "ON", the semi-streaming capabilities to none.

3, set the output phase current

In order to drive different torque stepper motors, the user can switch SW1 on the drive panel, SW2, SW3 bit to set the drive's output phase current (RMS) amperes, each switch position corresponding output current, different models drive different values of the corresponding output current. Detailed in Table 2.

Output current(A)				
SW1	SW2	SW3	PEAK	RMS
on	on	on	2.8	2.0
off	on	on	3.5	2.5
on	off	on	4.2	3.0
off	off	on	4.9	3.5
on	on	off	5.7	4.0
off	on	off	6.4	4.6
on	off	off	7.0	5.0
off	off	off	7.8	5.6

Table 2

4, semi-streaming capabilities

Semi-flow function means no step pulse 200ms, the driver output current is automatically reduced to 70% of rated output current is used to prevent motor heat.

四、Power Interface

1, VAC: Connect the drive power

AC and DC common.

2, A + A-B + B-: connect the two-phase hybrid stepping motor

Connect the drive and two-phase hybrid stepping motors using four-wire system, the motor windings in parallel and series connection, parallel connection, high-speed performance, but the drive current is large (1.73 times the motor winding current),
When connected in series with the drive current is equal to the motor winding currents.

五、Installation

Have 20mm of space around, can not be placed next to other heating equipment, to avoid dust, oil mist, corrosive gas, too much humidity and strong vibration

六、Trouble shooting

1, status lights indicate

RUN: green, normal work light.

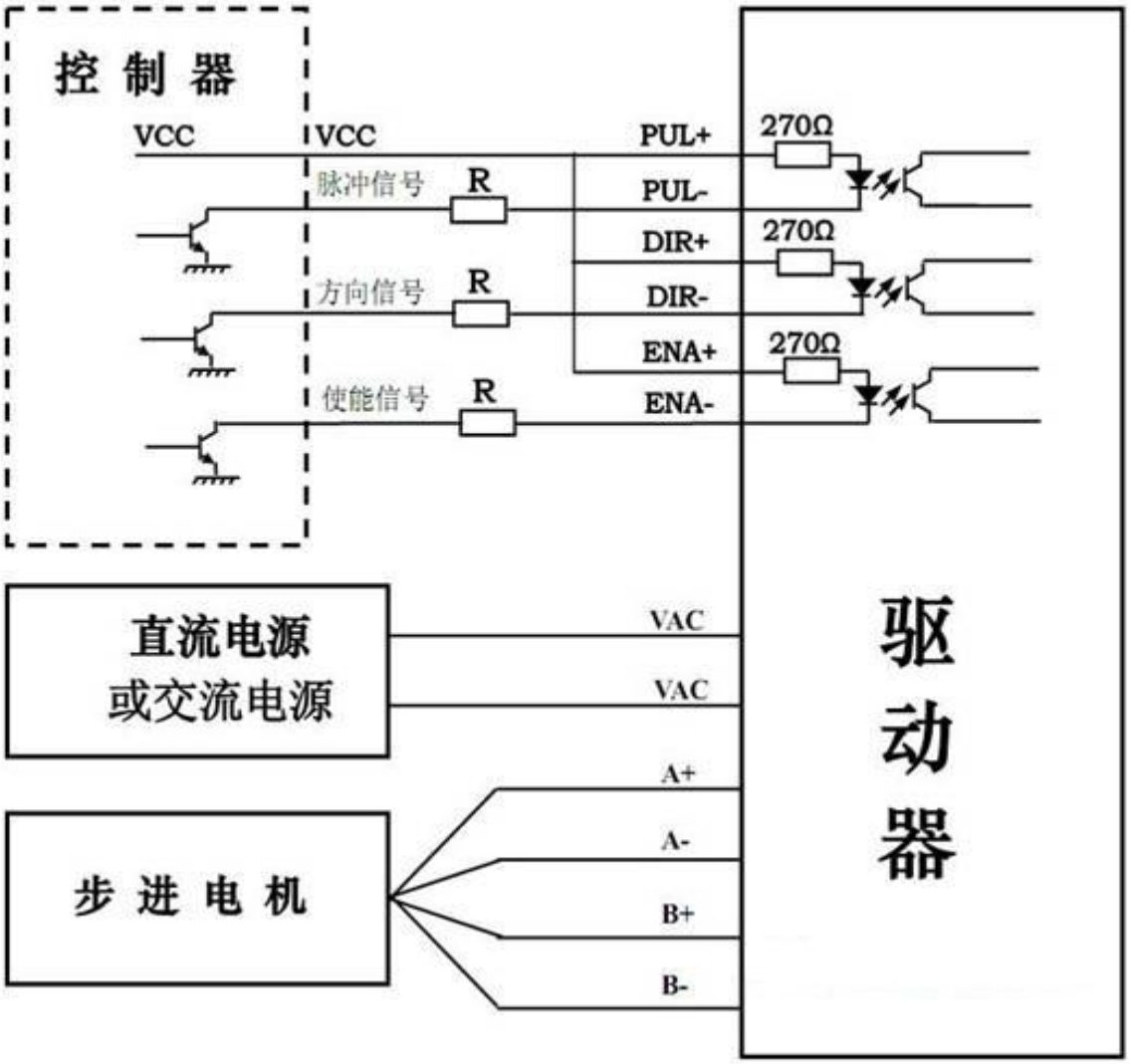
ERR: red, light failure, the motor phase short circuit, overvoltage and under voltage protection.

2、Trouble and Eliminate

Trouble	Reason	Solution
LED is not lit	Pick the wrong power	Check the power connection
	Low power supply voltage	Improve the supply voltage
Motor does not turn, and no holding torque	Wrong motor connection	Correct motor connections
	Offline enable RESET signal is active	Make RESET invalid
Motor does not turn, but holding torque	No pulse signal input	Pulse width and level adjustment signal
Motor rotates in the wrong direction	Power line phase sequence is wrong	Swap any two connected lines
	Wrong direction signal input	Change direction setting
Motor torque is too small	Phase current is too small	set Phase current correctly
	Too fast acceleration	Reduce the acceleration values
	Motor stall	Ruled out mechanical failure
	The drive does not match the motor	Change the appropriate drive

七、drive wiring

A complete stepper motor control system should contain stepper drives, DC power supply and controller (pulse source). The following is a typical system wiring diagram:


控 制 器
VCC
VCC
脉冲信号
R
方向信号
R
使能信号
R
PUL+
PUL-
DIR+
DIR-
ENA+
ENA-
270Ω
270Ω
270Ω
直流电源
或交流电源
VAC
VAC
步 进 电 机
A+
A-
B+
B-
驱
动
器